

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** CE

Commented [U1]: Please enter the name in Arial 12 font

2. **Subject Name:** Matrix Structural Analysis **L-T-P:** 2-0-2 **Credits:**3

Commented [U2]: Please enter the details (use Arial 12 font)

3. **Pre-requisites:** 1)Structural Analysis

Commented [U3]: Please enter the pre-requisites (subject name and subject no. in Arial 12 font)

4. **Syllabus and reference books:**

Commented [U4]: 1.Please type the syllabus and reference books in the box provided below.

Syllabus: Solution of algebraic equations, direct stiffness method, analysis of 2D structures (theory and computer implementation), analysis of 3D structures (theory and computer implementation), approximate methods for indeterminate trusses and industrial frames

Please use "Arial" font with a font size of "12" and a line spacing of 1.0 for syllabus and reference books. Please use Justified Text (Ctrl+J) for the content entered in the boxes
Reference books should be numbered as 1), 2), 3)
Please provide all bibliographic details for the reference books

Reference Books:

- 1) Matrix Structural Analysis by Gere and Weaver (Text book)
- 2) Matrix Structural Analysis by W McGuire, RH Gallagher and RD Ziemian (Reference book)
- 3) Matrix and Finite Element Analysis of Structures by M Mukhopadhyay and AH Sheikh (Reference book)
- 4) Computer Analysis of Framed Structures by D Maity (Reference book)

5. **Lecture-wise break-up:**

Commented [U5]: Please provide lecture-wise break-up in the table given below:

Sl. No.	Topic	No. of Hours
1.	Solution of algebraic equations: linear and nonlinear equations, implementation	6
2.	Direct stiffness method: concept, element stiffness equations; element flexibility equations; flexibility-stiffness transformation, global stiffness equations, the framework of element stiffness matrix, rotation of member axis, oblique and elastic supports, temperature effects, support settlement, non-prismatic and curved members	12
3.	Analysis of 2D structures (theory and computer implementation): plane trusses and frames. Implementation for oblique and elastic supports, temperature effects, support settlement, nonprismatic and curved members	12
4.	Analysis of 3D structures (theory and computer implementation): space trusses and	12

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	frames. Implementation for rotation of member axis, member principal axes in general direction, oblique and elastic supports, temperature effects, support settlement, non-prismatic and curved members	
5.	Approximate methods: Indeterminate trusses and industrial frames; analysis of building frames to gravity and lateral loads	6
Total number of hours		48

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